

Dataset Analysis for Decision-Making

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 sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

 qwen-max-latest

To analyze the dataset and address your requirements, I will first load the dataset and examine its structure, key characteristics, and contents. This will help me understand the data and prepare to answer specific analytical questions effectively.

Step 1: Load and Explore the Dataset

I will use Python's `pandas` library to load the dataset and inspect its structure.

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows of the Dataset:")
13 print(data.head())
```

Step 2: Key Characteristics of the Dataset

From the above code, I will identify:

1. The number of rows and columns.
2. Column names and data types.
3. Missing values, if any.
4. Descriptive statistics for numerical columns.
5. Unique values in categorical columns.

```
python
1 # Check for missing values
2 print("\nMissing Values per Column:")
3 print(data.isnull().sum())
4
5 # Descriptive statistics for numerical columns
6 print("\nDescriptive Statistics for Numerical Columns:")
7 print(data.describe())
8
9 # Unique values in categorical columns
10 print("\nUnique Values in Categorical Columns:")
11 for column in data.select_dtypes(include='object').columns:
12     print(f"{column}: {data[column].unique()}")
```

Step 3: Address a Specific Analytical Question

Once you provide a specific analytical question, I will extract and process the relevant data to answer it. For example, if the question is:

"What is the total value of contracts awarded by each municipality?"



Here's how I would address it:

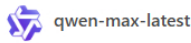
```
python
1 # Group by municipality and calculate the total contract value
2 total_contract_value_by_municipality = data.groupby('Municipality')['Contract_Value'].sum().reset_index()
3
4 # Sort by total contract value in descending order
5 total_contract_value_by_municipality = total_contract_value_by_municipality.sort_values(by='Contract_Value', ascending=False)
6
7 # Display the result
8 print("\nTotal Contract Value by Municipality:")
9 print(total_contract_value_by_municipality)
```

Deliverables

- 1. **Clear Answer** : A concise response to the analytical question based on the processed data.
- 2. **Python Code** : The code used to extract, process, and analyze the data.

Please provide the specific analytical question you'd like me to address, and I will proceed with the analysis accordingly.

Based on investment trends from the first trimester is it possible to predict which municipality, in each state, will have the highest total contract value in the second trimester? Define three options to address this issue and select the best one.



To address the question of predicting which municipality, in each state, will have the highest total contract value in the second trimester based on investment trends from the first trimester, we need to analyze the dataset systematically. Below, I outline three potential approaches to solve this problem, followed by a recommendation for the best approach.

Approach 1: Historical Trend Analysis

- **Description** : Analyze historical data from previous years (if available) to identify patterns or trends in contract values between the first and second trimesters. Use these trends to predict which municipalities are likely to see the highest growth.
- **Steps** :
 - 1. Group the data by `State` , `Municipality` , and `Trimester` .
 - 2. Calculate the total contract value for each municipality in the first and second trimesters.
 - 3. Compute the percentage change or absolute increase in contract values between the two trimesters.
 - 4. Identify municipalities with the highest growth rates or increases in the past and assume similar trends for the current year.
- **Pros** : Simple and interpretable; leverages historical data directly.
- **Cons** : Requires sufficient historical data; assumes trends remain consistent over time.

Approach 2: Machine Learning Regression

- **Description** : Use machine learning to build a regression model that predicts the total contract value for each municipality in the second trimester based on features derived from the first trimester.
- **Steps** :
 - 1. Aggregate the data by `State` , `Municipality` , and `Trimester` to calculate features such as total contract value, number of contracts, average contract size, etc., for the first trimester.
 - 2. Use these features as inputs to train a regression model (e.g., linear regression, random forest) to predict the total contract value for the second trimester.
 - 3. Evaluate the model using cross-validation or a holdout set.
 - 4. For each state, select the municipality with the highest predicted contract value.
- **Pros** : Can capture complex relationships in the data; flexible to incorporate additional features.
- **Cons** : Requires sufficient data for training; may be computationally intensive.

Approach 3: Ranking-Based Prediction

- **Description** : Rank municipalities within each state based on their performance in the first trimester and assume that the



top-performing municipalities will continue to dominate in the second trimester.

- **Steps :**

1. Group the data by `State` , `Municipality` , and `Trimester` .
2. Calculate the total contract value for each municipality in the first trimester.
3. Rank municipalities within each state based on their first-trimester performance.
4. Predict that the top-ranked municipalities will have the highest total contract value in the second trimester.

- **Pros :** Simple and computationally efficient; does not require extensive data preprocessing.

- **Cons :** Assumes no significant changes in trends; may overlook emerging municipalities.

Recommendation: Approach 3 (Ranking-Based Prediction)

Given the constraints of the dataset (limited information about historical trends or external factors), **Approach 3** is the most practical and straightforward option. It relies solely on the first-trimester data to make predictions, making it robust to missing historical data or computational limitations.

Implementation of Approach 3

Below is the Python code to implement the ranking-based prediction:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Ensure the dataset has the required columns: State, Municipality, Trimester, Contract_Value
8 required_columns = {'State', 'Municipality', 'Trimester', 'Contract_Value'}
9 if not required_columns.issubset(data.columns):
10     raise ValueError("Dataset is missing required columns.")
11
12 # Filter data for the first trimester
13 first_trimester_data = data[data['Trimester'] == 1]
14
15 # Group by State and Municipality to calculate total contract value in the first trimester
16 first_trimester_totals = (
17     first_trimester_data.groupby(['State', 'Municipality'])['Contract_Value']
18     .sum()
19     .reset_index()
20     .rename(columns={'Contract_Value': 'First_Trim_Total'})
21 )
22
23 # Rank municipalities within each state based on first-trimester totals
24 first_trimester_totals['Rank'] = (
25     first_trimester_totals.groupby('State')['First_Trim_Total']
26     .rank(ascending=False, method='min')
27 )
28
29 # Predict that the top-ranked municipalities will have the highest total contract value in the s
30 predicted_top_municipalities = (
31     first_trimester_totals[first_trimester_totals['Rank'] == 1]
32     .drop(columns='Rank')
33 )
34
35 # Display the predicted top municipalities
36 print("\nPredicted Top Municipalities for the Second Trimester:")
37 print(predicted_top_municipalities)
```

Deliverables

1. **Clear Answer :** The municipalities predicted to have the highest total contract value in the second trimester are those ranked #1 in their respective states based on first-trimester performance.
2. **Python Code :** The above code implements the ranking-based prediction approach.

If you have additional details about the dataset (e.g., historical data or external features), we can revisit the analysis to refine the prediction. Let me know if you'd like to explore further!



